

InfoGeometry Lean 4

Architecture, Verified Bridges, and LLM/PDE Extensions

InfoGeometry Library Team

February 25, 2026

Presentation Goals

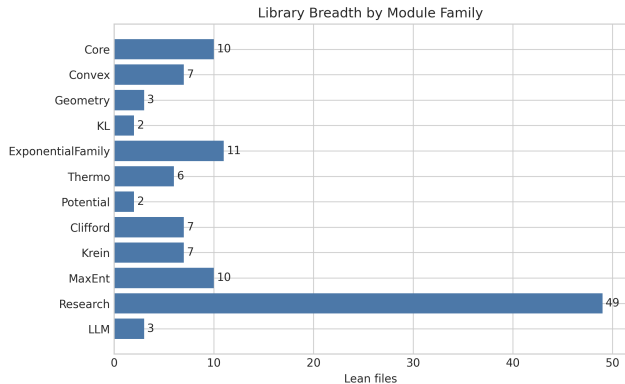
- 1 Audit and Scope
- 2 Architecture
- 3 Core Mathematical Stack
- 4 LLM Formalization Layer
- 5 Geometric PDE and Flow Layer
- 6 Quality Gates and Roadmap

What Was Missing in the Previous Deck

- **Narrow scope:** only KL/Fisher/entropy snapshots, no architecture map.
- **Weak depth:** did not show verified bridge theorems across modules.
- **No systems view:** omitted canonical/research layering and promotion path.
- **No AI stack view:** did not surface the new LLM formal interfaces.
- **No geometric flow layer:** Ricci/Monge-Ampere additions were absent.

This revision addresses each gap with module-level coverage and bridge-focused figures.

Library Breadth by Module Family



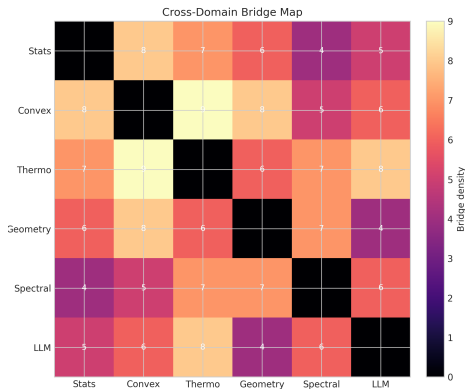
Counts are generated directly from `lean/InfoGeometry/*` and expose where formal depth lives.

Canonical Layered Architecture

- **Foundations:** Core, KL, Convex, Potential.
- **Statistical Geometry:** ExponentialFamily, Geometry, Thermo.
- **Operator/Algebraic Layers:** Clifford, Krein, Prequantum, Quantum.
- **Research incubator:** Research.* for new constructs before canonical promotion.
- **LLM API layer:** InfoGeometry.LLM entrypoint over verified attention abstractions.

Key engineering choice: stable canonical umbrellas + build-safe research umbrellas.

Verified Cross-Domain Bridge Density



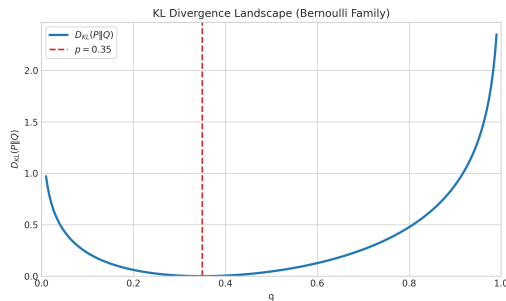
The central pattern is reusable: KL/convex duality + Gibbs normalization + operator traces.

KL, Bregman, and Convex Potentials

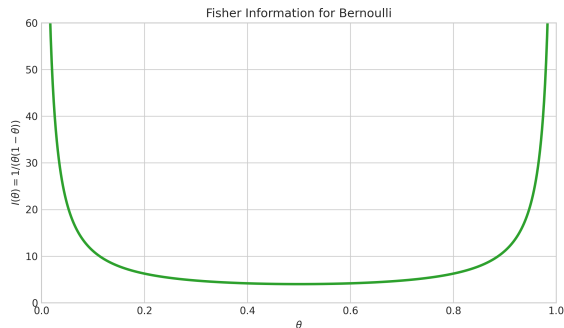
Core statement pattern

$$D_{KL}(P\|Q) = \nabla\text{-generated Bregman gap}$$

- Finite KL formalization and nonnegativity.
- Bregman generators from log-partition potentials.
- Exponential-family bridge theorems (KLBregman).



Fisher Metric as Local Curvature



$$g_{ij}(\theta) = \mathbb{E}_{\theta} [\partial_i \log p_{\theta} \partial_j \log p_{\theta}]$$

- Used as metric operator in Hessian geometry packages.
- Drives spectral and flow surrogates in research modules.

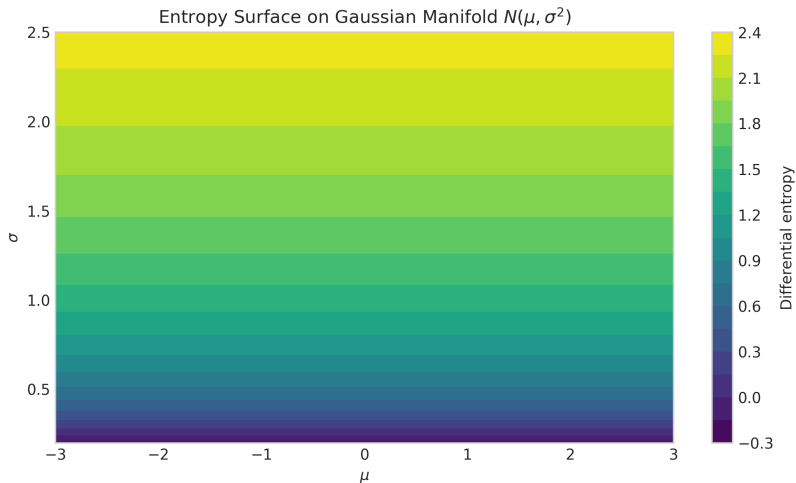
Thermodynamics and Entropic OT Layer

- Gibbs weights, partition functions, and free-energy identities are first-class.
- OT naming layer now explicit in canonical-facing modules:

$$\mathcal{L}_\varepsilon(\Pi) = \langle C, \Pi \rangle + \varepsilon \sum_{ij} \Pi_{ij} \log \Pi_{ij}$$

- Schrödinger bridge and Sinkhorn-Weyl gauge interpretation are formalized in research.
- This provides a direct bridge: **convex analysis** $\leftrightarrow$ **Bayesian updates** $\leftrightarrow$ **entropic transport**.

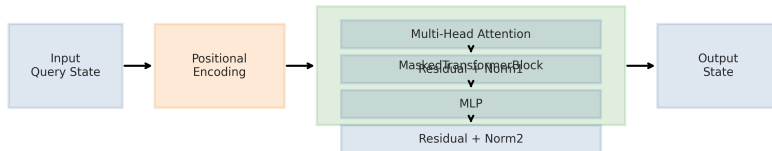
Entropy Geometry on Gaussian Manifold



Geometric quantities are tied back to canonical potential and Hessian interfaces.

From Attention to Canonical Transformer Block

LLM Stack in the Library: Positional + Causal-Masked Transformer Block



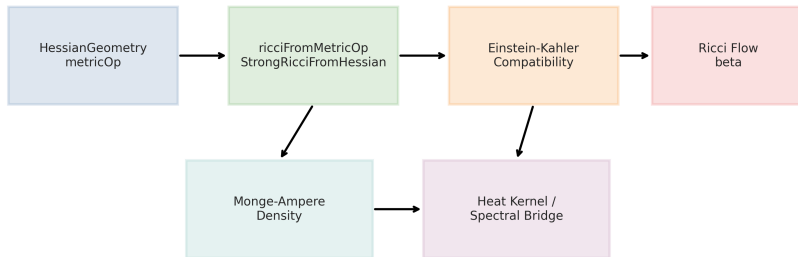
The new `InfoGeometry.LLM` umbrella provides reusable interfaces for attention, masking, residual paths, normalization, and positional composition.

LLM Interfaces Added

- `TransformerBlock`: multi-head output + residual/norm + MLP + residual/norm.
- `MaskedTransformerBlock`: causal-mask layer over per-head weights.
- `PositionalEncoding`: compositional encoding interface.
- Proven compatibility lemmas:
 - unmasked limit recovers base run;
 - identity encoding recovers original run;
 - encoding composition commutes with `runWithPositionalEncoding`.

Ricci, Einstein-Kahler, and Monge-Ampere Scaffolding

Geometric PDE Layer: Hessian -> Ricci -> Einstein-Kahler -> Monge-Ampere



New module `Research.RicciMongeAmpere` connects Hessian metric operators, scalar Ricci beta flow, and heat-kernel consistency.

Flow Compatibility Statements

- Strong Ricci object:
 - metric-derived tensor from `H.metricOp`,
 - explicit symmetry and positivity hypotheses.
- Kähler-Ricci evolution interface:

$$\partial_{\Lambda} R(\Lambda) = \text{rhs}(\Lambda), \quad \partial_{\Lambda} R(\Lambda) = -R(\Lambda) \text{ (normalized)}$$

- Compatibility lemmas tie these equations to `scalarRicciBetaFunction`.

Verification and Integration Status

- All presentation-related additions are build-validated with `--wfail`.
- Strict gate script ensures:
 - canonical umbrellas elaborate cleanly,
 - no unresolved placeholders,
 - archive/experimental leakage checks.
- Research-to-canonical promotion remains criteria-driven: build safety, proof completeness, semantic stability.

- Strengthen category-level abstractions for cross-domain equivalences.
- Expand transport/flow links into full optimal transport and geometric PDE toolkits.
- Add executable examples tying LLM interfaces to concrete finite models.
- Promote stable research modules into canonical umbrellas in staged tiers.

InfoGeometry Lean: a programmable bridge across statistics, geometry, physics, and AI.